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Cloud-based business services innovation: A risk management model

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ABSTRACT

Cloud computing providers and their software-as-a-service offerings have become more profuse and mature, making cloud technology an increasingly important platform for business services innovation. Although the cloud offers rich opportunities for transforming businesses—innovating existing services and introducing creative new ones—it also involves risks that business managers must identify and resolve to successfully drive innovation efforts. However, no comprehensive models are available to help managers assess and mitigate the risks they face. To address this void, we reviewed extant literature on cloud computing from a business innovation perspective to uncover the myriad challenges managers confront as they seek to leverage cloud technology in the ongoing transformation of their organization's service offerings. Combining this systematic literature analysis with relevant theory, we synthesized an integrated model for managing risk during innovation of cloud-based business services. The model identifies three types of risks (services, technology, and process risks) and four types of resolutions (stakeholder engagement, technology development, innovation planning, and innovation control). The model also helps managers identify their organization's general risk profile and link that profile to a specific configuration of resolutions.

1. Introduction

As a technology-enabling platform, cloud computing is increasingly seen as a major driver for business innovation. Although cloud computing is disruptive and can adversely affect companies (Werfs, Baxter, Allison, & Sommerville, 2013), once adopted, it offers organizations considerable innovation opportunities (Marston, Li, Bandyopadhyay, Zhang, & Ghalsasi, 2011; Willcocks, Venters, & Whitley, 2013). Cloud not only provides computing capability through large pools of automated, scalable computing resources (Cafaro & Aloisio, 2011; Venters & Whitley, 2012) and associated applications (Cusumano, 2010; Venters and Whitley, 2012), it also inspires cloud-based creativity by letting organizations use open source computing resources from third-party application providers to augment the established cloud platform. Through this ecosystem, organizations can outsource their technological needs and flexibly select the required prepackaged software capabilities to ignite their business outcomes. As such, cloud technology providers give organizations access to a rich pool of instruments to architect innovative solutions that both meet their needs and lower the barriers to innovation (Marston et al., 2011).

Our research specifically focuses on how an organization can leverage Software-as-a-Service (SaaS) to deliver cloud-based business services to its customers. SaaS delivers application functionality and related services through the Internet (Sultan, 2010; Low et al., 2011). Here, we view *innovation* as an organization's ability to develop and realize an idea that addresses a specific challenge and achieves value for its businesses or customers. Hence, cloud-based business services offer innovative solutions from a service provider to its customers, enabled by the cloud.

Cloud-based business services innovation presents a multiplicity of risks for the providing organization. Venters and Whitley (2012) posit that the cloud severely affects an organization's IT structure and interfaces, introducing key risks related to identity management, governance, compliance, software isolation, security responses, and so on (Ratten, 2016). Still, no comprehensive models exist to help managers assess and mitigate the considerable risks they face. Our research addresses this void by analyzing and synthesizing extant literature to offer a model of how cloud-based business services providers can manage the risks involved in innovating services for their customers.

2. Literature background

To position and frame our research, we review the literature on cloud-based business service innovation and offer key concepts from risk management theory.

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2.1. Cloud-based business service innovation

According to Sean Marston and his colleagues, cloud computing is

> "an information technology service model where computing services (both hardware and software) are delivered on-demand to customers over a network in a self-service fashion, independent of device and location. The resources required to provide the requisite quality-of-service levels are shared, dynamically scalable, rapidly provisioned, virtualized and released with minimal service provider interaction" (Marston et al., 2011; p. 177).

Cloud technology encompasses public, private, and hybrid models. It represents a paradigm shift enabled by both technological maturity and the shift toward service-centric offerings. The first strand of cloud computing emerged from technological innovations such as virtualization, high-performance networks and data center automation (Armbrust et al., 2010; Boss, Malladi, Quan, Legregni, & Hall, 2007; Venters & Whitley, 2012). The second strand emerged from an increasing emphasis on service-based perspectives (Etro, 2009; Vouk, 2008; Venters and Whitley, 2012). A distinctive feature of cloud services is the pay-as-you-use attribute.

Although cloud-based services come in the form of Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS), the focus of this research is SaaS, which delivers application functionality and related services through the Internet (Sultan, 2010; Low et al., 2011). Hence, as Fig. 1 illustrates, we are concerned with how cloud-based business services providers manage the risks incurred as they rely on cloud providers to innovate SaaS for their customers.

The original driver behind cloud computing for both large and small-to-medium enterprises (SMEs) was mainly the desire to outsource supporting tasks and focus on core business outcomes. They thus shifted attention from managing technology assets to considering the customer value in using technology services (Grönroos, 2011; Venters and Whitley, 2012). As such, cloud services based on the pay-as-you-go model give businesses a flexible budgetary framework to outsource their technological needs while they focus other resources on enriching business capabilities. Venters and Whitley (2012) articulate what businesses desire from the cloud in terms of *equivalence* (technical services that are at least equivalent and *abstraction* (technical services that abstract away unnecessary complexity), as well as scalability, efficiency, creativity, and simplicity.

As cloud providers and cloud offerings have become more profuse and mature, the cloud has developed into a platform that drives creativity. Cloud computing's distinctive nature offers many possibilities for innovating business (Willcocks et al., 2013). Cloud-based creativity arises from an ecosystem that leverages open source computing resources, with third-party application providers complementing the cloud platform and offering a comprehensive set of tools and building blocks for architecting desirable end-to-end business outcomes. Cloud providers give organizations access to a rich pool of instruments to architect innovative solutions for their customers, thereby lowering the barriers to innovation (Marston et al., 2011).

However, this complex environment of cloud-based business services innovation introduces a host of risks for cloud-based business services providers, their partners, and their customers. From the customer's perspective, the decision to shift from on-promise to the cloud could be precarious. Moreover, Venters and Whitley (2012) posit that shifting to the cloud influences intra-organizational interfaces. Key cloud computing risks for organizations relate to identity management, governance, compliance, software isolation, and security responses (Ratten, 2016). As such, the cloud's impact goes beyond IT and affects the entire business (Werfs et al., 2013). Jan Damsgaard, Kettinger, and Lacity (2014) propose that managers stop thinking about how cloud services can transform the IT function and instead ponder how cloud services can transform the entire business. The decisions entailed have potentially dire consequences. Furthermore, Marston et al. (2011) add to this burden by suggesting that business leaders proactively develop an overall strategy and timeline regarding which applications they can move to the cloud (Prasad & Green, 2015).

In summary, there is an emergent body of knowledge vis-à-vis cloud computing from a business perspective that goes beyond the initial focus on the technology's adoption and implementation. However, as Table 1 shows, none of the existing literature reviews covers risk management for cloud-based business services innovation. Cloud-based business services providers are clearly in dire need of models that can help them, their business partners, and other stakeholders address the host of challenges they face during service innovation.

2.2. Risk management theory

Risk management uses various heuristics to link risks and resolutions. In this context, *risk* refers to the probability and impact of an adverse outcome (Aven & Renn, 2009; Graham, Weiner, & Benesh-Weiner, 1995); *resolution* refers to resources and efforts to avoid, transfer, prevent, mitigate, or assume the risks (Nogueira & Bhattacharya, 2000); and the *heuristics* link specific risks to one or more appropriate resolutions (Iversen, Mathiassen, & Nielsen, 2004). Fig. 2 illustrates this general risk management approach. At any point in time, an organization's managers identify and evaluate the risks that might occur, their likelihood, and their potential impact. They then explore and evaluate which resolutions, if applied, would avoid, transfer, prevent, mitigate, or help them assume the risks that might be encountered in the future.

As Table 2 shows, in the information systems and software context, Iversen et al. (2004) specify four approaches to risk management: risk list, risk-action list, risk-strategy model, and risk-strategy analysis. The risk list approach offers a prioritized list of risks, starting with the highest risk. The approach is easy to build, modify, and work with. However, its major drawbacks are that it provides neither risk resolutions nor strategic oversight. The risk-action list approach is thus superior in that it provides risk resolutions for each risk item. However, the risk-action list approach shares one common drawback with the risk list approach: it does not offer strategic oversight, which is crucial for flexibility and adaptability. While the risk-strategy model approach is hard to build and modify, it offers managers this strategic oversight. Moreover, in contrast to the risk-strategy analysis approach, it is easy for managers to use. We therefore chose to adopt this approach as the foundation for building our risk-strategy model.

3. Research method

With a focus on cloud computing, risk management, business services, and innovation, we used qualitative literature review followed by model development as our research method. Asking how managers of cloud-based business services innovation can manage risks, we began a systematic review of related literature. We searched journals, covering the intersection of literature on SaaS, risk management, and innovation.

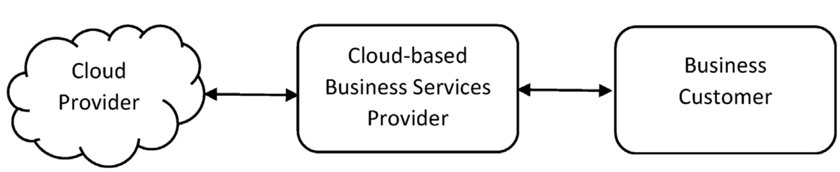

Fig. 1. Cloud-based Business Services Context.

Table 1
Extant Cloud Computing Literature Reviews in Business Context.

Title	Authors	Journal	Year
"A Descriptive Literature Review and Classification of Cloud Computing Research"	Haibo Yang and Mary Tate	*Communications of the Association for Information Systems*	2009
"A Critical Review of Cloud Computing: Researching Desires and Realities"	Will Venters and Edgar A. Whitley	*Journal of Information Technology*	2012
"Characteristics of Cloud Computing in the Business Context: A Systematic Literature Review"	Mar Stieninger and Deitmar Nedbal	*Global Journal of Flexible Systems Management*	2014
"Determinant Factors of Cloud-Sourcing Decisions: Reflecting on the IT Outsourcing Literature in the Era of Cloud Computing"	Stephan Schneider and Ali Sunyaev	*Journal of Information Technology*	2016

This yielded an initial list of 250 academic papers. We then manually scanned the titles and abstracts of these papers to narrow down the list to 31 papers that could provide insights into risk management in the context of cloud business services innovation (these papers are marked with an asterisk in our list of references).

Next, we engaged in systematic data analysis of the 31 papers to identify and categorize text passages relating to cloud-based business services innovation risks, resolutions, and heuristics. First, we used open coding to identify passages in the texts that describe relevant phenomena (Myers, 2013). As part of that process, we recorded and categorized all text passages relating to cloud-based business services innovation risks and resolutions. We documented a total of 260 passages related to risks and 270 related to resolutions. We then used axial coding to conceptualize and group the passages into 20 risk items and 35 resolution items. *Axial coding* is a qualitative research method that specifies each category's properties and dimensions (Charmaz, 2014). To ensure rigorous analysis, the third author initially trained the first and second authors as coders on literature subsets until they were able to achieve an 85 percent or above intercoder reliability (Mouter & Vonk Noordegraaf, 2012).

Finally, we adapted McFarlan's (1981) framework of risk management to conceptualize our categories into three overarching risk areas and four overarching resolution areas, and then linked the risk areas to the appropriate resolution areas. McFarlan's framework is robust and is extensively cited in the literature. It offers a flexible and strategic approach to managing risk in IT projects. McFarlan's (1981) risk areas are: project size, in terms of the amount of resources needed; the development team's experience with technology; and the project structure, which reflects how well the task is understood. The resolution areas are external integration, which links the project team to other stakeholders; internal integration, which ensures that the team operates as an integrated unit; formal planning to schedule the tasks; and formal control, which focuses on the post-deployment phase.

Adapting these concepts to cloud-based business services innovation corresponds to selective coding, which selects core categories and relates them to other categories (Strauss & Corbin, 1990). This adaptation process allowed us to distinguish three overarching risk areas (business services risks, technology risks, and process risks) and four resolution areas (stakeholder engagement, technology development, innovation planning, and innovation control).

Our research was thus guided by an interpretative philosophical stance in which reality is viewed as socially constructed. As such, we attempted to understand phenomena through the meanings that researchers assigned them in their publications, and we defined *quality* as it relates to the plausibility of our overall argument (Myers, 2013). We were sensitive to possible biases and systematic distortions in the narratives collected from the literature and participants (Myers, 2013), as well as our own. To mitigate these biases, we scrutinized data analysis and conclusions among the research team members. To further increase the rigor of our study, we shared our conclusions and the resulting risk-strategy model with five IT executives of cloud-based business services. Their roles varied from technical to managerial, and each had more than 15 years of experience in IT services. We sent these executives our overview, problem statement, risk model, and summary via email, and followed up with a one-hour face-to-face or phone session with each executive. We then analyzed and synthesized the feedback to inform and fine-tune the final model.

4. Conceptual development

In this section, we present the synthesized concepts about risks and risk resolutions drawing on McFarlan's (1981) theoretical framework.

4.1. Innovation risk areas

Our review revealed a plethora of risks related to cloud-based business services innovation. As noted previously and shown in Table 3, we categorized these risks into three risk areas as follows. *Services risks* correspond to McFarlan's (1981) notion of the problem's structure and relate to the involved actors' understanding of the intended cloud-based business service. *Technology risks relate* to the involved actors' skills and understanding of the involved technology (McFarlan, 1981). *Process risks* correspond to McFarlan's (1981) notion of project size and relate to the complexity and sophistication level of the innovation processes.

4.1.1. Services risks

The evolution of the cloud, frequent introduction of new technologies, and deeper focus on data analytics has produced opportunities and

Fig. 2. Risk Management Framework.

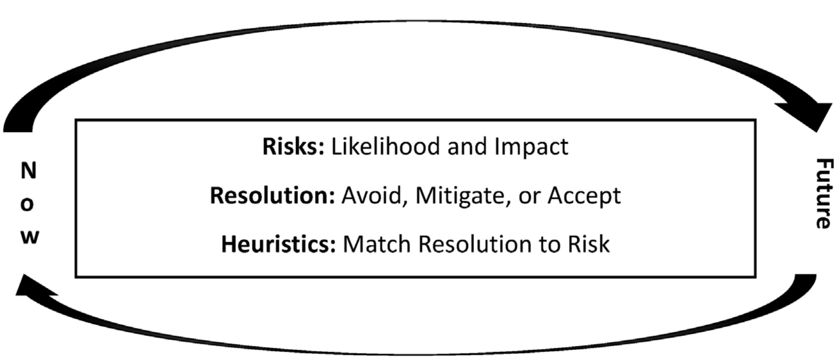

Table 2
Types of Risk Management Approaches (Iversen et al., 2004).

Type of Approach	Characteristics	Assessment	Exemplars
Risk list	A list of prioritized risk items	+ Easy to use + Easy to build + Easy to modify + Risk appreciation − Risk resolution − Strategic oversight	(Barki, Rivard, & Talbot, 1993; Keil, Cule, Lyytinen, & Schmidt, 1998; Moynihan, 1996; Ropponen and Lyytinen, 2000
Risk-action list	A list of prioritized risk items with related resolution actions	+ Easy to use + Easy to build + Easy to modify + Risk appreciation + Risk resolution − Strategic oversight	(Alter & Ginzberg, 1978; Boehm, 1991; Jones 1994; Ould, 1999)
Risk-strategy model	A contingency model that relates aggregate risk items to aggregate resolution actions	+ Easy to use − Easy to build − Easy to modify + Risk appreciation + Risk resolution + Strategic oversight	(Donaldson & Siegel, 2001; Keil et al., 1998; McFarlan, 1981)
Risk-strategy analysis	A stepwise process that links a detailed understanding of risks to an overall risk management strategy	− Easy to use − Easy to build + Easy to modify + Risk appreciation + Risk resolution + Strategic oversight	(Davis, 1982; Mathiassen, Munk-Madsen, Nielsen, & Stage, 2000)

challenges for cloud-based business services providers. One of these challenges is compatibility with a customer's products and services; the cloud-based business service must align with customers' desired business outcomes, policy, and environment if they are to accept that service as a solution (Lin & Chen, 2012; Lacity and Reynolds, 2014).

From a financial perspective, a cloud-based business service's return on investment (ROI) could be long term, and focusing solely on costs can entail risks (Venters & Whitley, 2012). Still, management must minimize cost and acquire the ability to predict costs associated with cloud-based business services market-entry (Kaltenecker, Hess, & Huesig, 2015; Werfs et al., 2013). Cost saving is one of the primary reasons that organizations switch to cloud computing, but these expected savings are not always realized. Further, whether targeting long- or short-term returns, not all managers view cloud-based business services innovation as an investment. One reason for this might be that cloud-based business services require time for market visibility, awareness, acceptance, and standardization. And, while most customers can see the benefits of cloud-based business services innovation, not all are willing to immediately invest in business services innovation (Werfs et al., 2013). Indeed, some companies will not pursue cloud-based business services innovation at all until the uncertainties—such as the security and standardization concerns, and the lack of successful business models—are addressed (Lin & Chen, 2012).

From a legal perspective, in today's globalized economy, the cloud tends to be widely distributed across multiple domestic and international jurisdictions. When a cloud-based business service is offered across such diverse geographies, a set of unique challenges emerges and the jurisdictional uncertainties about data protection law may dissuade businesses (McAlpine, 2010; Ward & Sipior, 2010). Moreover, businesses could be exposed to lawsuits related to the accidental release of data or to sharing infrastructure, platform, software, and data resources with other customers of a cloud provider (Harris & Alter, 2010; Ward & Sipior, 2010).

4.1.2. Technology risks

Technology risks pose other challenges due to the speed of change characterizing today's business environment. Technological costs include those associated with training internal resources such as software developers, business development managers, and the sales team. Moreover, as customers innovate their cloud-based business services, they, too, must climb a learning curve. Hence, the requisites to build, harvest, and share knowledge are important considerations for cloud-based business services innovation (Lin & Chen, 2012).

In addition, dependence on cloud providers makes it hard to control the service level agreements (SLAs) for customers. Moreover, the encryption and decryption mechanisms that many cloud services use might slow the response times (Lacity & Reynolds, 2014) and the network dependency could impede access to cloud services as they evolve. These challenges may be outside an organization's sphere of control (Battleson, West, Kim, Ramesh, & Robinson, 2016). Further, compared to on-premises software, customers have far higher expectations from on-demand software regarding its attributes, including its accuracy and robustness (Kaltenecker et al., 2015).

The major technological concerns here relate to identity management, governance, compliance, software isolation, and security responses (Lacity & Reynolds, 2014). As nearly every survey on cloud services shows (Lacity & Reynolds, 2014), organizations want cloud-based business services to provide added security and privacy measures as the innovation is being introduced (Lin & Chen, 2012). Addressing security concerns is vital because the perception of a low level of security can negatively affect customers attitude toward new cloud-based business services (Arpaci, Kilicer, & Bardakci, 2015).

Because of their natural off-premise placement, cloud-based business services can lack transparency and the services provided may be poorly understood. Services are abstracted in a way that makes it harder for customers to manage their IT (Venters & Whitley, 2012). This absence of transparency is mainly due to insufficient information about the processing operation (Seddon & Currie, 2013). Moreover, integration with the supporting cloud technology tends to be complex and poorly understood (Oliveira, Thomas, & Espadanal, 2014). When looking at the underlying infrastructure, many cloud service providers are not compatible with each other (Stieninger & Nedbal, 2014). Additionally, providers are challenged by the need to integrate flexibility into their service delivery, while shielding customers from the underlying complexity (Cheng, Yang, Akella, & Tang, 2011).

Table 3
Risk Areas and Items.

Risk Area	Risk Definition	Risk Items
Service Risks	Drawing on McFarlan's (1981) notion of problem structure, these risks relate to the current understanding of the intended cloud-based business service.	1. The cloud-based business service is misaligned with the customer's desired business outcomes, policy, or environment (Lin & Chen, 2012; Kaltenecker et al., 2015; Lacity & Reynolds, 2014; Stieninger & Nedbal, 2014). 2. The cost associated with the cloud-based business service market entry is high or unpredictable (Kaltenecker et al., 2015; Werfs et al., 2013). 3. The cloud-based business service is offered across diverse geographies with unique challenges, such as legal issues (Ward & Sipior, 2010; Lacity & Reynolds, 2014; Seddon & Currie, 2013). 4. The ROI of the cloud-based business service is long term (Battleson et al., 2016; Venters & Whitley, 2012). 5. The cloud-based business service targets markets that require time for visibility, awareness, acceptance, and standardization (Lin & Chen, 2012; Kaltenecker et al., 2015; Werfs et al., 2013).
Technology Risks	Drawing on McFarlan's (1981) notion of the supporting technology, these risks relate to available skills and the technology's features.	1. Costs are associated with training internal staff members (development and sales) and customers (usage) on new cloud-based business services (Lin & Chen, 2012; Battleson et al., 2016; Kaltenecker et al., 2015). 2. Dependence on cloud providers makes it hard to control cloud-based business service SLAs for customers (Venters & Whitley, 2012; Seddon & Currie, 2013; C & tinean & Cândea, 2013; Stieninger & Nedbal, 2014; Lacity & Reynolds, 2014; Kaltenecker et al., 2015; Battleson et al., 2016; Werfs et al., 2013). 3. Cloud-based business services require added measures to address security and privacy concerns (Lin & Chen, 2012; Arpaci et al., 2015; Ward & Sipior, 2010; Lacity & Reynolds, 2014; Stieninger & Nedbal, 2014). 4. Basing the business service on cloud technology leads to a lack of transparency and poor understanding of the nature of the service provided (Venters & Whitley, 2012; Seddon & Currie, 2013). 5. The cloud-based business service may not be thoroughly tested and might rely on immature technology (Kaltenecker et al., 2015). 6. The supporting cloud technology is complex and poorly understood (Lacity & Reynolds, 2014; Cho et al., 2009; Kaltenecker et al., 2015; Fogarty, 2009; Stieninger & Nedbal, 2014).
Process Risks	Drawing on McFarlan's (1981) notion of project size, these risks relate to the complexity and sophistication level of the innovation processes.	1. All key stakeholders are not fully committed to the cloud-based business service (Lacity & Reynolds, 2014; Eaton et al., 2014). 2. It is challenging to transition from on-premise to cloud-based business services (Battleson et al., 2016; Oliveira et al., 2014; Lin & Chen, 2012). 3. The supporting cloud technology's provider has not paid sufficient attention to the innovation of this business service (Lacity & Reynolds, 2014) 4. The cloud-based business service's customers do not have full control over requisite data (Seddon & Currie, 2013; Lin & Chen, 2012) 5. The organization's current IT governance approach does not align well with realizing the business value associated with cloud computing services (Block, 2012; Prasad et al., 2014) 6. Additional costs exist related to running business service applications remotely (Loebbecke & Huyskens, 2006; Schneider & Sunyaev, 2016; Brook et al., 2014)

4.1.3. Process risks

Cloud-based business services providers deal with many stakeholders, including investors, internal management, the cloud provider, and customers. The main stakeholders are not always fully committed to the cloud-based business service. This gives rise to risk factors, including the decision makers' attitude toward outsourcing, cost uncertainty, the fear of losing control, and the customer's strategic vulnerability. Further, gaining commitment and support from all internal and external stakeholders can also be challenging (Lacity & Reynolds, 2014). Finally, providers and customers may avoid innovation as it is a tradeoff with stability and existing investments (Lin & Chen, 2012).

An additional process risk is that customers may no longer have direct control over their data and thus cannot exercise technical and managerial measures to ensure access, integrity, confidentiality, and transferability of their data as they endeavor to innovate (Seddon & Currie, 2013). When cloud-based business services customers embark upon innovation, they may not have full control over the requisite data. This can delay projects, while skill gaps and time pressures make project development uncertain (Lin & Chen, 2012). Such uncertainty arises when providers of supporting cloud technology pay insufficient attention to the innovation of a business service (Lacity & Reynolds, 2014).

Finally, the provider's IT governance approach to cloud-based business services innovation may not be well aligned with the customer's values (Block, 2012; Prasad, Green, & Heales, 2014). Additionally, running business service applications remotely can give rise to other issues (Brook, Feltkamp, & van der Meer, 2014; Loebbecke & Huyskens, 2006; Schneider & Sunyaev, 2016).

4.2. Innovation resolution areas

Risks management requires that organizations identify preventative measures and mitigating actions, or decide which of the risks to accept. Our literature review revealed insights on ways to manage risk in cloud-based business services innovation. We identified four resolution areas. *Stakeholder engagement* involves the business service owners, operators, and partners that are part of the innovation effort, corresponding to McFarlan's (1981) notion of external integration. *Technology development* focuses on technological skills and capabilities, drawing on McFarlan's (1981) notion of internal integration. *Innovation planning* is concerned with the innovation process, including the tasks, how to sequence them, and how to allocate necessary resources; this resolution area draws on McFarlan's (1981) notion of formal planning. Finally, *innovation control* focuses on monitoring, evaluating, and course-correcting the innovation process, corresponding to McFarlan's (1981) notion of formal control.

4.2.1. Stakeholder engagement

Customers are the ultimate stakeholders, and management must know their needs, issues, and internal processes (Cho, Cheng, & Hung, 2009; Kaltenecker et al., 2015). Cloud-based business services must fit customers' business models and addresses their functional and non-functional requirements (Lin & Chen, 2012; Werfs et al., 2013). Providers must therefore be far more consumer-centric and manage the customers' expectations and perceptions, which differ from those in traditional on-premise situations (Kaltenecker et al., 2015). With the customer in mind, service attributes must be aligned with the targeted market and reconciled across different geographies (Brook et al., 2014; Kaltenecker et al., 2015; Seddon and Currie, 2013; Werfs et al., 2013).

Cloud-based business services innovation requires collaboration. Before launching, the cloud-based business services should be thoroughly tested. Hence, the service must be designed to be testable with associated partners (Lin & Chen, 2012; Kaltenecker et al., 2015; Rogers, 1995; Sabi, Uzoka, Langmia, & Njeh, 2016). Further, as new market entrants, service providers need an effective marketing strategy to boost their footprint. Established traditional on-premise vendors transitioning to cloud-based business services can capitalize on their established brand names (Kaltenecker et al., 2015; Sabi et al., 2016).

4.2.2. Technology development

Cloud-based business services providers must build adequate managerial, technical, and relational capabilities (Bharadwaj, 2000; Battleson et al., 2016; Dyer and Singh, 1998; Garrison, Wakefield, & Kim, 2015; Lacity & Reynolds, 2014; Oliveira et al., 2014; Ravichandran et al., 2005). When offering new cloud-based business services, the required human capital can be acquired externally and by engaging internal staff members who are excited and committed (Kaltenecker et al., 2015). To reduce the cost of acquiring expertise, resources such as students and virtual workforces may be employed (Kaltenecker et al., 2015).

A service integration development platform should be built to allow development of a set of rich and diverse cloud-based business services (Cheng et al., 2011; Lin and Chen, 2012). From a customer perspective, innovative services must be compatible with the existing environment to allow a seamless progression (Lin & Chen, 2012) and the functionally should be packaged into smaller services to reduce initial launching costs (Battleson et al., 2016; Werfs et al., 2013). A hybrid cloud solution offers a sound option for customers who want to maintain control over their sensitive data and applications by deploying them in their private cloud, while benefiting from public cloud characteristics for other services (Venters & Whitley, 2012).

4.2.3. Innovation planning

Planning strategies include step-wise transformation, inside-out drive, and expert, leader, trial-and-error, and spin-off strategies (Kaltenecker et al., 2015; Prasad et al., 2014). Moreover, organizations must secure the necessary financial resources, with a focus on long-term ROI (Battleson et al., 2016; Kaltenecker et al., 2015).

Offering incentives to top management can help promote and successfully execute an innovative vision of cloud-based business services (Kaltenecker et al., 2015; Prasad and Green, 2015). Additionally, management should consult with the development team and view the team as a key stakeholder for success. Focusing on human capital ensures that the technology infrastructure is built adequately to allow integration of the cloud-based solution (Cheng et al., 2011; Oliveira et al., 2014; Eaton, Hallingby, Nesse, & Hanseth, 2014; Sabi et al., 2016). Further, with the focus on long-term quality, all products must be designed to be cloud-enabled (Kaltenecker et al., 2015) and, in these complex ecosystems, it is essential to select strong collaborative partners and cloud providers (Battleson et al., 2016; Cheng et al., 2011; Eaton et al., 2014; Kaltenecker et al., 2015; Prasad et al., 2014; Venters and Whitley, 2012; Whitley & Willcocks, 2011; Ward & Sipior, 2010).

4.2.4. Innovation control

During the innovation process, management must continuously inspire the developers and the sales personnel to maintain high-quality operational offerings. To align human capital, managers can continuously communicate the vision and the directions in which to build, thereby instilling these key elements in the corporate culture (Kaltenecker et al., 2015). It is a good practice to follow a complementary product strategy that is compatible with the existing values, past experiences, and needs of potential adopters (Kaltenecker et al., 2015; Lin and Chen, 2012). To remain competitive, cloud-based business services providers must monitor the big players and price their own offerings accordingly (Kaltenecker et al., 2015).

Because it is a top customer concern, security requirements must be addressed early. The cloud-based service should use only cloud providers with the required security certifications. Such certifications let business owners evaluate and monitor the cloud provider's performance (Battleson et al., 2016; Oliveira et al., 2014; Sabi et al., 2016; Werfs et al., 2013). To ensure growth opportunities, cloud-based business services providers can predesign cloud-based business services to be customizable to segments beyond the current market (Eaton et al., 2014). Adjustments to market dynamics and shifting needs must be balanced, however, with a persistent overall strategy (Eaton et al., 2014; Kaltenecker et al., 2015; Prasad et al., 2014; Werfs et al., 2013). Hence, appropriate governance structures must be put in place to provide this balance between the strategy and the market dynamics (Brook et al., 2014; Prasad et al., 2014; Prasad and Green, 2015).

5. Risk management model

Using (McFarlan, 1981) as a theoretical lens, our model supports three risk management activities: assessment of the risks (see Tables 3 and 4), identification of appropriate strategy (Table 5), and planning the risk mitigation (Table 6). The following sections describe how to use our risk-strategy model following these basic activities.

5.1. Assess the risks

As Table 3 shows, the first activity is to assess each risk area. The *service risk area* relates to the actors' understanding of the intentions of the cloud-based business services innovation. Services risks include ROI and alignment with the customer's business outcome, expenditure, and geographic and legal issues. The *technology risk area* relates to available skills and the supporting technology's features. Technology risks include the expenditure associated with training, customers' SLA challenges, security concerns, lack of product quality, and the technological solution's overall complexity. The *process risk area* relates to the complexity and sophistication of the innovation process and includes challenges related to securing stakeholders' commitment, hurdles in the transition to cloud-based business services, the lack of cloud provider operational support, inadequate governance structures, and costs associated with being off-premise.

To evaluate each risk area, Table 4 provides key questions and suggested answers. Managers can score each question between low (1) and high (5) and aggregate the individual scores of underlying risk items into an overall score for each overarching risk area. Based on the detailed assessments and the overall score, managers can then qualitatively assess the score for each risk area to be either high or a low; this yields the risk profile for each of the three risk areas.

5.2. Identify a strategy

In the second activity, managers use the first activity's result to select the corresponding risk profile in Table 5. For each risk profile, the risk-strategy model suggests the appropriate emphasis—high, medium, or low—for each of the four resolution areas: stakeholder engagement, technology development, innovation planning, and innovation control.

Table 4
Risk Assessment.

	Risk Assessment	Low Risk = 1	High Risk = 5	Score
Services Risks	How well do the cloud-based business services align with the customers' desired business outcomes, policies, and environment?	The cloud-based business services need no modifications to meet the customer's desired business outcomes, comply with the customer's policies, and fit the customer's environment.	The cloud-based business services require major modifications to meet the customer's desired business outcomes, comply with the customer's policies, and fit the customer's environment.	
	How predictable are the cost factors associated with market entry, and how much do they cost?	The cloud-based business services provider is already a market leader and the market entry factors are predictable and low cost.	The company is a new entrant and the market entry factors are unpredictable and expensive.	
	To what extent is the cloud-based business service part of a globally distributed network?	The cloud-based business service solution is limited to a single legal jurisdiction.	The cloud-based business service solution crosses several geographies with diverse legal jurisdictions.	
	How easy is it to realize a positive ROI?	The ROI measures are well known and are expected to be realized quickly.	The ROI measures are uncertain and are expected to take a long time to realize.	
	To what extent does the target market trust that the cloud-based business service adds benefits?	The cloud-based business services have a high market acceptance and are already standardized.	The cloud-based business service is new with no demonstrated added benefits.	
	Total Score (between 6 and 30)			
Technology Risks	How much training do the cloud-based business service personnel need?	The managers, the technical team, and the sales force have extensive experience with cloud-based business services.	The managers, the technical team, and the sales force have wide capability gaps related to the cloud-based business services.	
	How supportive is the cloud infrastructure in ensuring the customer's SLAs?	The connectivity to the cloud is reliable and the cloud features are scalable.	The connectivity to the cloud is unreliable and the cloud features are not scalable.	
	How well do the infrastructure security measures support the customer's requirements?	The customer's security requirements are supported and the cloud infrastructure has appropriate security certification.	The customer's security requirements are not supported and the cloud infrastructure has no appropriate security certification.	
	How well specified are the technical features of the cloud-based business service specified?	The cloud-based business service's technical features are specific and understood by the customers.	The cloud-based business service's technical features are abstract and poorly understood by the customers.	
	How robust is the cloud-based business service?	The cloud-based business service is thoroughly tested and is based on mature technology.	The cloud-based business service is not thoroughly tested and relies on immature technology.	
	How complex is the cloud-based business service?	The supporting clouds are compatible, and the cloud-based business services are intuitive and easy to deploy.	The supporting clouds are heterogeneous, and the cloud-based business services require considerable effort to explain and deploy.	
	Total Score (between 5 and 25)			
Process Risks	How committed are the stakeholders to the cloud-based business service?	The cloud-based business service's top management is engaged and committed to the long-term investment, while the customers are fully committed to innovating their on-premise services.	The cloud-based business service's top management is detached and uncommitted to the long-term investment; customers are hesitant to outsource their on-premise services.	
	How much effort is needed to transition to the cloud-based business service?	The integration with the customer's current platforms, processes, and workflow is easy and seamless.	The integration with the customer's current platforms, processes, and workflow is complicated and disruptive to business.	
	How attentive are the cloud providers to the cloud-based business services provider?	The cloud providers are responsive to the cloud-based business services provider, and contractual clauses enforce change requests.	The cloud providers are indifferent to the cloud-based business services provider, and no contractual clauses exist to enforce change requests.	
	How much control do the customers require over requisite data?	The customers require no control over the requisite data, which is readily accessible.	The customers require substantial control over requisite data, which is difficult to acquire.	
	To what extent is the customer's IT governance optimized to maximize the business value associated with cloud-based business services?	The customers have substantial experience with cloud-based business services and require no adjustments in how they manage IT provisioning arrangements.	The customers have no experience with cloud-based business services and must significantly modify how they manage IT provisioning arrangements.	
	What is the additional cost related to running business service applications remotely?	The customers do not require technical specificity, so there is no additional cost related to running business service applications remotely.	The customers require substantial technical specificity, resulting in significant additional cost related to running business service applications remotely.	
	Total Score (between 6 and 30)			

Table 5
Linking Risk Profiles to Resolution Patterns.

Risk Type				Innovation Approach			
Profile	Service Risk	Technology Risk	Process Risk	Stakeholder Engagement	Technology Development	Innovation Planning	Innovation Control
1	Low	Low	High	Low	Medium	High	High
2	Low	Low	Low	Low	Low	Medium	High
3	Low	High	High	Low	High	Medium	Medium
4	Low	High	Low	Low	High	Low	Low
5	High	Low	High	High	Medium	High	High
6	High	Low	Low	High	Low	Medium	High
7	High	High	High	High	High	Low	Low
8	High	High	Low	High	High	Low	Low

Managers can then map the risk profile to a risk resolution strategy that emphasizes the target resolution areas.

5.3. Plan the risk mitigation

Specific resolution actions apply resources and effort to avoid, transfer, prevent, mitigate, or assume risks (Nogueira and Bhattacharya, 2000Nogueira & Bhattacharya, 2000); activity two's recommended risk resolution strategy tells managers where to dedicate the most resources and effort, and where to spend less. Following this logic, Table 6 offers exemplar actions for each resolution area to help managers identify specific actions to best mitigate the risks they face. Generally, managers should apply the appropriate actions with an appropriate level of focus. They can also choose other actions that suit their specific situation and circumstances.

6. Discussion

Cloud-based business services provide a natural platform and vast potential for business innovation (Willcocks et al., 2013). Innovation is imperative; organizations face ever-increasing customer demands for improved service offerings and reduced total cost of ownership. To grow and remain competitive, many of these organizations choose the cloud as their innovation platform. Against this backdrop, our research focuses on how organizations can manage the risks involved in cloud-based business services innovation as they deliver SaaS to their customers (Sultan, 2010; Low et al., 2011). In this context, we view innovation as developing and realizing an idea to address a specific challenge and achieve value for both the organization and its customers.

Businesses are in the cloud to obtain equivalence, abstraction, scalability, efficiency, creativity, and simplicity. Several authors (Marston et al., 2011; Willcocks et al., 2013; Werfs et al., 2013) argue that a key advantage here is that the move offers new and significant opportunities to disrupt and transform organizations because cloud technology gives organizations access to a rich pool of solutions that lower IT barriers to innovation and business transformation (Damsgaard et al., 2014; Marston et al., 2011). However, innovation of cloud-based business services involves a multiplicity of services, technology, and process risks. Hence, effective risk management continues to be critical for organizations choosing to innovate in the cloud for growth and competitiveness. Accordingly, our systematic literature review and resulting risk-strategy model contribute important insights and tools for risk management in the domain of cloud-based business services innovation.

Some literature studies seek to synthesize the insights of the growing body of knowledge on cloud technology (Stieninger & Nedbal, 2014; Venters & Whitley, 2012; Schneider & Sunyaev, 2016; Yang & Tate, 2009). Research and literature in the area of risk management is also readily available. However, none of these reviews focus on synthesizing the extant body of knowledge in the domain of cloud-based innovation of business services from a risk management perspective. We therefore engaged in a systematic review of risk management insights in the cloud-based business services innovation literature. The resulting contribution synthesizes this literature for risk management purposes.

Although the literature argues that cloud technology offers new opportunities for innovating business services (Werfs et al., 2013; Willcocks et al., 2013; Marston et al., 2011; Damsgaard et al., 2014), it also makes clear that businesses face severe risks as they increasingly leverage cloud technology to enable innovation of their services (Lin & Chen, 2012; Kaltenecker et al., 2015; Lacity and Reynolds, 2014; Marston et al., 2011; Venters and Whitley, 2012; Werfs et al., 2013). We found that there are no concepts and models available that managers can use to understand, assess, and mitigate the risks they face as they seek to exploit cloud technology for business service innovation. Our model offers a conceptualization of the risks into three areas: services risks (Lin & Chen, 2012; Kaltenecker et al., 2015; Werfs et al., 2013); technology risks (Battleson et al., 2016; Cătinean & Cândea, 2013; Kaltenecker et al., 2015; Lacity and Reynolds, 2014; Seddon & Currie, 2013; Stieninger & Nedbal, 2014; Venters & Whitley, 2012; Werfs et al., 2013); and process risks (Battleson et al., 2016; Oliveira et al., 2014; Lin & Chen, 2012). For each of these risk areas, we have synthesized risk items from extant literature.

Similarly, our model conceptualizes risk resolutions for innovation of cloud-based business services in four areas: stakeholder engagement (Lin & Chen, 2012; Rogers, 1995; Kaltenecker et al., 2015; Sabi et al., 2016); technology development (Cheng et al., 2011; Lin and Chen, 2012; Werfs et al., 2013; Battleson et al., 2016); innovation planning (Battleson et al., 2016; Cheng et al., 2011; Eaton et al., 2014; Kaltenecker et al., 2015; Prasad et al., 2014; Venters and Whitley, 2012; Whitley & Willcocks, 2011; Ward & Sipior, 2010); and innovation control (Eaton et al., 2014; Kaltenecker et al., 2015; Prasad et al., 2014; Werfs et al., 2013). Similar to the risk areas, each resolution area is grounded in the extant body of knowledge with exemplary resolution actions from the literature.

To help managers respond to risks as they seek to innovate through cloud-based business services, we synthesized risks and resolutions into a comprehensive model with heuristics for how resolutions apply to risk areas (Table 5). This risk-strategy model (Iversen et al., 2004) provides managers with holistic insight and an easy-to-use approach that recommends which strategies to apply for specific risk profiles. Hence, as a contribution to theory, we have extended McFarlan's (1981) risk management framework beyond software projects to cloud-based business services innovation. The framework targets eight risk profiles and links each profile to a particular configuration of risk resolution areas. As an important contribution to practice, we explicated the three risk management activities in cloud-based business service innovation: assess the risks, identify a strategy, and plan the risk mitigation. For each of these, we offered practical tools (Tables 3–6) that allow cloud-based business services providers to manage both internal risks and risks in their relations to cloud providers and the businesses they serve (Fig. 1).

Table 6
Resolution Areas and Actions.

Resolution Area	Definition	Resolution Actions
Stakeholder Engagement	Drawing on McFarlan's (1981) notion of external integration, these resolutions represent activities related to engaging business service owners, operators, and partners in the innovation effort.	• Survey customer needs, issues, and internal processes (Kaltenecker et al., 2015; Cho et al., 2009) • Manage customer expectations and perceptions (Kaltenecker et al., 2015) • Address customer nonfunctional requirements early (Werfs et al., 2013) • Provide cloud-based business services that fit the customer's business model (Lin & Chen, 2012) • Make the cloud-based business service testable with associated partners (Lin & Chen, 2012; Rogers, 1995; Kaltenecker et al., 2015; Sabi et al., 2016) • Use collaboration to support innovation (Kaltenecker et al., 2015) • Reconcile geographical and organizational approaches to trans-border data flow regulations (Seddon & Currie, 2013) • Align the cloud-based business services with the attributes of the target market (Werfs et al., 2013; Kaltenecker et al., 2015; Brook et al., 2014) • Capitalize on the cloud provider's brand name when offering the new services (Kaltenecker et al., 2015) • Use marketing to generate awareness and become visible in the established market (Kaltenecker et al., 2015; Sabi et al., 2016)
Technology Development	Drawing on McFarlan's (1981) notion of internal integration, these resolutions represent activities related to the development of technological skills and capabilities.	• Increase managerial, technical, and relational capabilities (Bharadwaj, 2000; Ravichandran et al., 2005; Dyer & Singh, 1998; Garrison et al., 2015; Battleson et al., 2016; Lacity & Reynolds, 2014; Oliveira et al., 2014) • Acquire established expertise when planning to offer new business services (Kaltenecker et al., 2015) • Engage internal personnel who are excited about the innovative cloud-based business services (Kaltenecker et al., 2015) • Utilize inexpensive resources, such as students and virtual workforces (Kaltenecker et al., 2015) • Build a service integration development platform to enable scaling to suit application diversity (Cheng et al., 2011 Lin & Chen, 2012) • Make the innovative cloud-based business services compatible with the existing development environment to ease adoption (Lin & Chen, 2012) • Package the functionality into smaller services to reduce initial costs (Werfs et al., 2013; Battleson et al., 2016) • Use redundant cloud services to reduce network dependency (Battleson et al., 2016)
Innovation Planning	Drawing on McFarlan's (1981) notion of formal planning, these resolutions represent activities related to planning the innovation process, including which tasks to emphasize and how to sequence them and allocate requisite resources.	• Adopt appropriate strategies such as step-wise transformation, inside-out drive, and expert, leader, trial-and-error, and spin-off strategies (Prasad et al., 2014; Kaltenecker et al., 2015); • Become nimble, efficient, and agile (Werfs et al., 2013) • Secure financial resources and focus on long-term ROI (Battleson et al., 2016; Kaltenecker et al., 2015) • Incentivize top management to execute an innovative vision of cloud-based business services (Kaltenecker et al., 2015; Prasad and Green, 2015) • Prepare on-premise products to be cloud-enabled (Kaltenecker et al., 2015) • Ensure that the technology infrastructure can integrate the cloud-based solution (Oliveira et al., 2014; Cheng et al., 2011; Eaton et al., 2014; Sabi et al., 2016) • Select successful and collaborative partners and cloud providers (Ward & Sipior, 2010; Kaltenecker et al., 2015; Prasad et al., 2014; Venters & Whitley, 2012; Whitley & Willcocks, 2011Battleson et al., 2016; Cheng et al., 2011; Eaton et al., 2014) • Utilize hybrid cloud solution to give business owners control over sensitive data and applications (Venters & Whitley, 2012) • Invest in liability insurance to mitigate potential legal challenges (Ward & Sipior, 2010)
Innovation Control	Drawing on McFarlan's (1981) notion of formal control, these resolutions represent activities related to monitoring, evaluating, and course-correcting the innovation process.	• Motivate developers and sales personnel (Kaltenecker et al., 2015) • Communicate directions to internal personnel (Kaltenecker et al., 2015) • Address security requirements and concerns early (Werfs et al., 2013) • Require security certifications so that business owners can evaluate and monitor the cloud provider's performance (Battleson et al., 2016; Oliveira et al., 2014; Sabi et al., 2016) • Monitor the big players and set pricing according to competitors (Kaltenecker et al., 2015) • Use complementary product strategy to be compatible with the existing values, past experiences, and needs of potential adopters (Kaltenecker et al., 2015) • Gain additional business payoffs by opening cloud-based business services capabilities to organizations beyond the current market (Eaton et al., 2014) • Follow the strategy with persistence and provide support during execution (Werfs et al., 2013; Eaton et al., 2014; Kaltenecker et al., 2015; Prasad et al., 2014) • Deploy appropriate governance structures (Prasad et al., 2014; Brook et al., 2014; Prasad & Green, 2015)

One area for future research is to explore whether the risk-strategy model works differently for public, private, and hybrid cloud environments. Our research did not differentiate between these categories; hence, further insights may be gained from investigating the differences and commonalities across these contexts. Another area for future studies is how cloud-based business services providers might deal with risks related to transitioning all or some services off the cloud. In other words, what goes on the cloud may very well have to come off some day. Researching this area is of great importance; this was clear in our interviews with experienced IT executives as part of our research. These practitioners also pointed out that, in addition to innovation risks, other risk groupings could also be investigated, including differential risks, contract risks, organizational risks, and reputational risks. Such a broader categorization of risks related to managing cloud-based business services could provide valuable insights.

7. Conclusion

In conclusion, we have contributed a literature synthesize for risk management in cloud-based business service innovation as an extension of McFarlan's (1981) theoretical framework. Moreover, we have provided practitioners with a risk-strategy model to manage risks in these contexts. It is important to stress that the model is based on the literature. Because of the pace of change in technology and related business architectures, some of the model's details might therefore need further development in the future. Still, with its solid anchoring in extant literature and its strategic risk orientation the model offers an adaptable and intuitive framework that will continue to be relevant for both theory and practice.

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